

Durham County Cellular Tower Zoning Summary (Unincorporated Areas)

Applicable Ordinance Section

Durham's Unified Development Ordinance (UDO) contains specific standards for **Telecommunications and Antennas**, particularly under *Sec. 5.3.3N – Wireless Communication Facilities (WCFs)*. These rules apply to new **freestanding cellular towers** (wireless support structures) in the unincorporated parts of Durham County. Below is a summary of the key requirements:

Height Restrictions by Zoning District

New cell tower heights are capped based on the zoning district and whether the tower is **concealed** (camouflaged or unipole design) or **non-concealed** (standard monopole with exposed antennas):

- **Rural (RR) District:** Maximum **120 feet** height for both concealed and non-concealed towers udo.durhamnc.gov.
- **Low-Density Residential (RS-20) District:**
 - Concealed towers may reach about **55 feet** (UDO allows up to 20 feet above the normal 35' building height limit) udo.durhamnc.gov.
 - Non-concealed towers are limited to **55 feet** udo.durhamnc.gov. (Note: Non-concealed towers in RS-20 and RR also require a **minimum lot size of 5 acres** udo.durhamnc.gov.)
- **Medium/High-Density Residential (RS-10, RS-8) Districts:**
 - Concealed towers may be allowed (approximately **20 ft above** the base building height limit) with special approval.
 - Non-concealed towers are **NOT permitted** in RS-8 or RS-10 zones udo.durhamnc.gov.
- **Planned Development Residential (PDR) District:** Treated like a residential zone – a concealed tower can be up to 20 ft above the district's building height, but any tower would require special use permit review. Non-concealed towers in PDR are generally subject to the same restrictions as residential zones (e.g. no setback reductions and likely special permit required).
- **Commercial, Industrial, and Other Non-Residential Districts:** Maximum **180 feet** height for both concealed and non-concealed towers udo.durhamnc.gov. In these districts, concealed towers

meeting standards may be permitted with a simpler process (see permitting below), whereas non-concealed towers still require special approval.

Summary of Height Limits: Concealed towers are allowed taller heights in residential areas than non-concealed. For example, in most residential districts the tallest concealed towers can be ~20 feet above the normal building height (often totaling ~50–60 ft)udo.durhamnc.gov, whereas non-concealed towers in residential areas are either sharply limited in height (55 ft in RS-20) or prohibited (in RS-8/RS-10)udo.durhamnc.gov. Rural and non-residential areas allow the greatest heights (up to 120' in rural and 180' in industrial/commercial) for new towersudo.durhamnc.gov.

Setbacks and Fall Zone Requirements

Setbacks: All new freestanding towers must observe significant setbacks from property lines to ensure a *fall zone*. The minimum setback from each property line must be **at least 120% of the tower's height or 85 feet, whichever is greater**udo.durhamnc.gov. This effectively means the tower's fall zone should be fully on-site with a buffer. For example, a 100' tower must be at least 120' from any property boundary.

- These setback rules apply in all cases (both concealed and non-concealed towers). In **residential districts** (including PDR), no reduction in the 120%/85' setback is allowed. In **non-residential districts**, an applicant may seek a **minor special use permit** to reduce the setback below 120% of height (if justified)udo.durhamnc.gov.
- **Pipeline Buffer:** If a tower is near a major natural gas transmission line ($\geq 8"$ diameter), it must also be set back at least 120% of the tower height or 85' from the **pipeline easement's centerline**, whichever is greaterudo.durhamnc.gov. This adds an extra safety buffer in such cases.

Fall Zone: By imposing the 120%-of-height setback, the ordinance creates an effective fall zone on the property so that if a tower were to collapse, it should fall within the site. For any proposed increase in height on an existing tower, the County may require an engineering analysis of the fall zone impacts with the taller structureudo.durhamnc.gov. (Generally, new tower applications must demonstrate that the tower will not pose danger to off-site structures in the event of failure.)

Zoning Districts – Where Towers Are Allowed

New freestanding cell towers are allowed in a range of zoning classifications in unincorporated Durham County, **with stricter controls in residential areas:**

- **Residential Zones:** Freestanding towers in residential and planned residential (PDR) zones are **heavily restricted**. Concealed towers (e.g. stealth/unipole designs) may be permitted **with a special use permit** in residential areas, but non-concealed towers face limits. In fact, as noted, non-concealed towers **cannot be built at all in RS-8 or RS-10** residential districts udo.durhamnc.gov. In the low-density RS-20 zone, a non-concealed tower is allowed only on a tract at least 5 acres in size udo.durhamnc.gov and with a height limit of 55' as above. Towers in any residential setting will require a public hearing and special use approval (see below).
- **Rural/Agricultural Zones (RR):** Freestanding towers are allowed in the Rural Residential district. Concealed or not, they can reach 120' in height udo.durhamnc.gov. A non-concealed tower in RR still needs a large parcel (≥5 acres) udo.durhamnc.gov and a special use permit. This zoning, being sparsely developed, is one of the more permissive for towers (subject to the setback and height limits).
- **Commercial and Industrial Zones:** New towers are generally **permitted** (often as a limited use) in commercial, office, research, and industrial districts, since these areas are intended to accommodate infrastructure. Concealed towers are typically allowed subject to the UDO's design standards (with administrative site plan approval) udo.durhamnc.gov, whereas non-concealed towers will require at least a minor special use permit. Most commercial/industrial zones allow up to 180' height udo.durhamnc.gov and have no minimum lot size for towers beyond the standard setbacks.
- **Design Districts/Overlays:** (Primarily within city jurisdiction) The question focuses on *unincorporated county*, so downtown or compact design districts are not applicable. However, note that towers near designated scenic roadways have special requirements regardless of zone (detailed under "Permitting" below).

Concealed vs. Non-Concealed: The UDO draws a distinction: "*Freestanding, Concealed (or Unipole) WCF*" versus "*Freestanding, Non-Concealed WCF*." A **concealed tower** means the antennas and equipment are camouflaged – for example, housed within a unipole (a pole with internal antennas) or disguised as a structure or tree. A **non-concealed tower** is a traditional cell tower with external antennas. Concealed towers are treated more leniently: they are allowed in more zones and often with a simpler approval path. By contrast, non-concealed towers face additional hurdles (such as the acreage requirement in residential zones and outright prohibition in the smallest residential zones) udo.durhamnc.gov.

Permitting and Public Hearing Process

Special Use Permits: In most cases, constructing a new freestanding cell tower will require obtaining a **special use permit (SUP)** in Durham County. The UDO distinguishes Minor vs. Major SUP:

- A **Minor Special Use Permit (mSUP)** (quasi-judicial) is required for **all new freestanding non-concealed towers** under normal circumstances udo.durhamnc.gov. This entails a public hearing before the *Durham Board of Adjustment*, where the applicant must present evidence to meet specific ordinance criteria. The Board of Adjustment can approve or deny the permit.
- A **Major Special Use Permit (MSUP)** is required if the tower is proposed in particularly sensitive situations. For example, **if a tower (even concealed) is proposed within 300 feet of a designated North Carolina Scenic Byway**, a Major SUP is mandated udo.durhamnc.gov. A Major SUP involves a public hearing and approval by the elected governing body (the Board of County Commissioners for projects in the County's jurisdiction) durhamnc.gov udo.durhamnc.gov. This higher level of review is intended to protect scenic vistas.

Permitted By Right (Limited Use): Some **concealed** WCFs that fully comply with UDO standards may be treated as **limited uses** ("L" in the use table) in certain non-residential districts engagedurham.com. A limited use is allowed without a special use permit, provided it meets all the preset conditions in UDO Sec. 5.3.3N. For instance, a concealed tower in an industrial zone that meets height, setback, and design standards could potentially proceed with administrative **Site Plan approval only** udo.durhamnc.gov. In such cases, no public hearing is required. Always, the **Section 5.1 Use Table** and notes determine whether a particular proposal in a given zone is Permitted by right, Limited (L) with conditions, or requires a Minor or Major SUP engagedurham.com.

Approval Criteria: When a special use permit is required, the applicant must demonstrate several findings beyond the general SUP findings. The UDO specifically requires evidence that **no feasible collocation alternative exists** – i.e. the applicant must show it is not reasonably possible to achieve the coverage/objectives by mounting on an existing tower or structure within the search area udo.durhamnc.gov udo.durhamnc.gov. If an existing tower or tall structure is available nearby, that should be used instead of building a new tower (this is a key policy in the ordinance). Other typical SUP considerations include compatibility with the area, minimal adverse visual impact, and compliance with all ordinance standards.

Public Hearing: Both Minor and Major SUP processes involve a formal public hearing (a quasi-judicial proceeding)durhamnc.gov. Neighbors are notified and given an opportunity to present evidence, and the decision-making body weighs the evidence against the ordinance criteria. This public hearing requirement ensures that new tower proposals are vetted for local impacts such as aesthetics, property values, and safety.

Appeals if Denied

If a tower permit application is denied at the administrative level (e.g. site plan not approved) or a special use permit is denied by the Board of Adjustment or Commissioners, the **appeal process** is as follows:

- **Administrative Decisions:** A denial of a site plan or similar administrative decision can typically be appealed to the Durham Board of Adjustment (as an “appeal of an administrative decision”) within 30 days. The Board of Adjustment would then hold a hearing to affirm, reverse, or modify the decision based on ordinance compliance.
- **Quasi-Judicial Decisions (SUP Denial):** If the Board of Adjustment denies a minor special use permit, or the Board of County Commissioners denies a major special use permit, the **only appeal is to the courts**. North Carolina law provides that appeals of quasi-judicial zoning decisions go to **NC Superior Court** (by filing a petition for writ of certiorari) within 30 days of the decision. There is no further appeal to any other local boarddurhamnc.gov. In other words, an applicant who is denied a tower SUP would have to seek judicial review; the courts will review whether the decision was made in accordance with the law and whether the findings were supported by competent evidence.

(Note: There is no “variance” possible for the use or height of a cell tower, since these are regulated uses; the Board of Adjustment cannot simply waive the use standards except through the SUP process or a hardship variance for unusual property conditions not related to the use.)

Permit Implementation and Expiration

After securing zoning approvals, the developer must obtain a **building permit** for the tower. Once a building permit is issued, state law governs its validity period. **North Carolina General Statutes §160D-1111** provides that a building permit will expire by limitation **6 months after issuance if work has not commenced**. If work has begun but then stops, the permit expires after a **12-month discontinuation** of workncleg.gov. In short, the tower’s construction must start within 6 months of the permit, and once started, progress should not halt for more than a year, or else a new permit would be required. Local

regulations do not shorten this timeframe (Durham follows the standard state building code timeline).

Additionally, Durham's inspections department may allow one extension (typically 6 months) upon request, but absent an extension, the above state deadlines apply. Once construction is actively underway, the building permit remains valid for the duration of the project as long as there are no lengthy pauses.

(Note: The zoning special use permit or site plan approval for the tower may have its own "sunset" clause – for instance, many development approvals expire if not acted upon within a certain period (often 2-3 years) – but the UDO text for WCFs does not specify a unique expiration, so general plan approval expirations would apply.)

Design and Safety Requirements

Durham's ordinance includes **additional standards** to ensure towers are constructed safely and with minimal visual intrusion:

- **Concealment and Aesthetics:** Towers should be **concealed** whenever possible. A "concealed" freestanding WCF means **all antennas and cabling are internal or camouflaged** so that the tower appears to be something else or simply a monopole with no visible arrays udo.durhamnc.gov. The UDO even defines criteria for **"monopine" or faux-tree towers**: to count as concealed, the fake tree must be placed in a cluster of real trees of a certain size, designed to resemble a species of tree in that area, with branches that cover at least the upper 50% of the pole, etc. udo.durhamnc.gov. If those criteria aren't met, a faux-tree tower would be treated as non-concealed udo.durhamnc.gov. These rules incentivize designs that blend with the environment (such as stealth flagpoles, church steeples, clock towers, or trees). Non-concealed towers (traditional monopoles with exposed panels) are effectively limited to more rural or industrial locations. Moreover, the ordinance requires **monopole construction** for all new towers – guyed or lattice towers are generally not permitted for new WCFs (monopoles are cleaner in appearance) udo.durhamnc.gov.
- **Co-Location Capacity:** All new towers must be built to accommodate multiple wireless carriers (to avoid the need for additional towers in the area). Specifically, towers under 120' must be engineered for at least **two** antenna arrays; towers 120–150' for at least **three** arrays; towers taller than 150' for at least **four** arrays udo.durhamnc.gov. This means space and structural capacity must be provided for other companies' equipment. The County also cannot deny a collocation that meets certain federal criteria (per Section 6409 of the

Spectrum Act), and collocations on existing towers are generally approved administratively in an expedited process. The ordinance's intent is clearly to **encourage collocation** and **minimize the proliferation of new towers**udo.durhamnc.gov. An applicant for a new tower is required to justify why collocating on any existing structure *will not work* before a new tower is approvedudo.durhamnc.govudo.durhamnc.gov.

- **Structural and Wind Load:** Towers must meet **structural standards** (ANSI/TIA engineering standards for wind loads, etc.) and the NC Building Code. The application must include documentation sealed by a professional engineer certifying the tower's structural integrity and code complianceudo.durhamnc.govudo.durhamnc.gov. This includes designing for Durham's wind speeds and any seismic considerations. The foundation, anchoring, and structure must support the antennas and co-location loading.
- **Lighting:** No lighting is allowed on a tower except what is required by the FAA. Any FAA-required obstruction lights must use the minimum intensity and number of flashes per minute permitted, and strobe lights are prohibited in favor of red beacons if allowedudo.durhamnc.gov. Lights must be oriented or shielded to avoid illuminating nearby residencesudo.durhamnc.gov. In practice, towers under 200' generally do not require FAA lighting; Durham caps height at 180', which often avoids lighting, but if terrain or airport proximity triggers FAA lighting, the tower will use dual red/white beacons with careful orientation.
- **Signs:** No commercial advertising signs are allowed on any tower or antenna structure. Only a small, informational sign is permitted at the base (for facility identification, owner contact, and safety warnings)udo.durhamnc.govudo.durhamnc.gov. This sign typically includes the tower owner's name, a 24-hour emergency phone number, and any federally required registration placard.
- **Security and Screening:** The base of the tower (the equipment compound) must be secured and screened. The UDO requires at minimum an **8-foot-tall opaque fence or wall** around the compoundudo.durhamnc.govudo.durhamnc.gov. Additionally, a **landscaped buffer** at least 10 feet wide with 60% opacity year-round must surround the compound to visually screen itudo.durhamnc.govudo.durhamnc.gov. Existing vegetation can count toward this buffer if preserved. These measures hide ground equipment and deter trespassing.

- **Radio Frequency Compliance:** Applicants must certify that the facility will comply with FCC radio-frequency (RF) exposure standards. While local governments cannot regulate RF emissions beyond federal law, the County does require documentation that the proposed installation **will not interfere with public safety communications** systems udo.durhamnc.gov. A report by an RFI expert must show that the new tower's transmissions won't cause harmful interference to the City/County emergency services radio network udo.durhamnc.gov (which operates in the 800 MHz spectrum). If interference were to occur, the tower operator must resolve it.
- **Insurance:** The tower owner must maintain at least **\$1,000,000 in general liability insurance** for the facility udo.durhamnc.gov. Proof of this insurance must be provided with the application and kept up to date on file. This ensures that any accidents or damage (for example, a tower collapse or falling ice debris) are financially covered.
- **Maintenance and Abandonment:** While not explicitly asked, it is worth noting that the ordinance (in line with typical standards) would require that towers be maintained in good condition. If a tower is ever abandoned or unused for a certain period (commonly 12 months), the County can require its removal. Any obsolete or unused tower must be removed by the owner to prevent derelict structures. (The UDO's WCF section includes procedures for removal of facilities no longer in use, consistent with FCC rules.)

Other Notable Provisions

- **Temporary Towers:** (Aside from the scope of "new freestanding" towers) The UDO allows temporary mobile communication towers under separate provisions (with a temporary use permit) for short-term events or emergencies udo.durhamnc.gov. These are not subject to the full zoning process if truly temporary.
- **Federal and State Preemptions:** The ordinance is crafted to comply with the federal Telecommunications Act and NC statutes. For example, time limits ("shot clocks") for permit decisions and allowances for certain modifications by right (the Spectrum Act 6409 collocation approvals) are acknowledged udo.durhamnc.gov udo.durhamnc.gov. Durham County cannot unreasonably discriminate among providers or effectively prohibit service under the Telecommunications Act, so the ordinance provides a path for approval in all zoning categories, with conditions to mitigate impacts.

Planning & Zoning Contact Information

For further information or guidance, you should contact the Durham City-County Planning Department (which administers zoning for unincorporated Durham County):

- **Durham City-County Planning Department** (Ground Floor of City Hall)
101 City Hall Plaza, Durham, NC 27701 durhamnc.gov
Phone: 919-560-4137 durhamnc.gov (main planning line)
Email: durhamnc.gov (for general inquiries to planning staff)
- **Zoning Administration Staff:** The Planning Department's Zoning Administration division can assist with specific ordinance interpretations and application procedures. They can be reached via the main line above.
- **Inspections/Building Permits:** Once zoning approval is obtained, building permits are handled by the Durham City-County Inspections Department. Their contact is 919-560-4144 (and they are at 101 City Hall Plaza as well). They can advise on building code requirements and permit application for the tower structure.

Always ensure to reference the latest Unified Development Ordinance provisions, as regulations can be amended. (This summary is based on the UDO as of 2024-2025, including text from **Section 5.3.3N – Wireless Communication**

Facilities udo.durhamnc.gov udo.durhamnc.gov and related sections.) For official determinations or if considering an application, a pre-submittal meeting with Planning staff is highly recommended. The Durham Planning staff can verify zoning district requirements for a specific site and guide you through the special use permit process if needed.

Sources:

- Durham City-County Unified Development Ordinance, Sec. 5.3.3N “*Wireless Communication Facilities (WCFs)*” (accessed via Durham UDO online) – setback, height, and permitting standards udo.durhamnc.gov udo.durhamnc.gov udo.durhamnc.gov udo.durhamnc.gov udo.durhamnc.gov.
- Durham UDO Sec. 5.3.3N.4 “*Standards for Specific WCFs*” – concealed vs. non-concealed tower criteria, co-location and design requirements udo.durhamnc.gov udo.durhamnc.gov udo.durhamnc.gov.
- Durham UDO Sec. 5.3.3N.3 “*General Requirements*” – lighting, signage, insurance, interference, and maintenance standards udo.durhamnc.gov udo.durhamnc.gov udo.durhamnc.gov.

- Durham UDO Sec. 5.1 “*Use Table*” (Latest version updated through Oct. 2024) – allowable districts and permit types for Wireless Communication Facilities engagedurham.com engagedurham.com.
- North Carolina General Statutes §160D-1111 – Expiration of Building Permits (applicable state law) ncleg.gov.
- Durham City-County Planning Department – Contact information durhamnc.gov.